

Section 08 - System Administrator Guide

What This Guide Covers

This guide explains how to deploy and operate SalesOps AI on AWS. It covers tool installation, credential management, backend deployment, first admin user promotion, frontend configuration, routine maintenance, and operational troubleshooting.

End-user instructions for representatives and business managers are in Section 07.

System Architecture Summary

SalesOps AI is a fully serverless application. All infrastructure is defined in `backend/template.yaml` and deployed with AWS SAM and CloudFormation.

Components created during deployment:

- Amazon API Gateway REST API, stage dev.
- AWS Lambda functions for auth, content, exam, dashboard, and release logic.
- Amazon Cognito User Pool and web app client.
- Four Amazon DynamoDB tables: `Users`, `Personas`, `Scenarios`, `ExamSessions`.
- Amazon SQS queue for delayed exam issue release.
- AWS Secrets Manager secret for the OpenAI API key.

The frontend is a Vite and React application. It connects to the deployed API Gateway endpoint through an environment variable.

Prerequisites

Required Tools

Install all tools before attempting deployment.

Tool	Minimum version	Purpose
Node.js	20.x	Frontend build and npm scripts
npm	10.x	Package management
AWS CLI	v2	AWS access and credential management
AWS SAM CLI	latest	SAM build and deploy

Verify tool versions before each deployment session:

```
node -v
npm -v
aws --version
sam --version
```

Install AWS CLI

Download from the official AWS docs. On macOS with Homebrew:

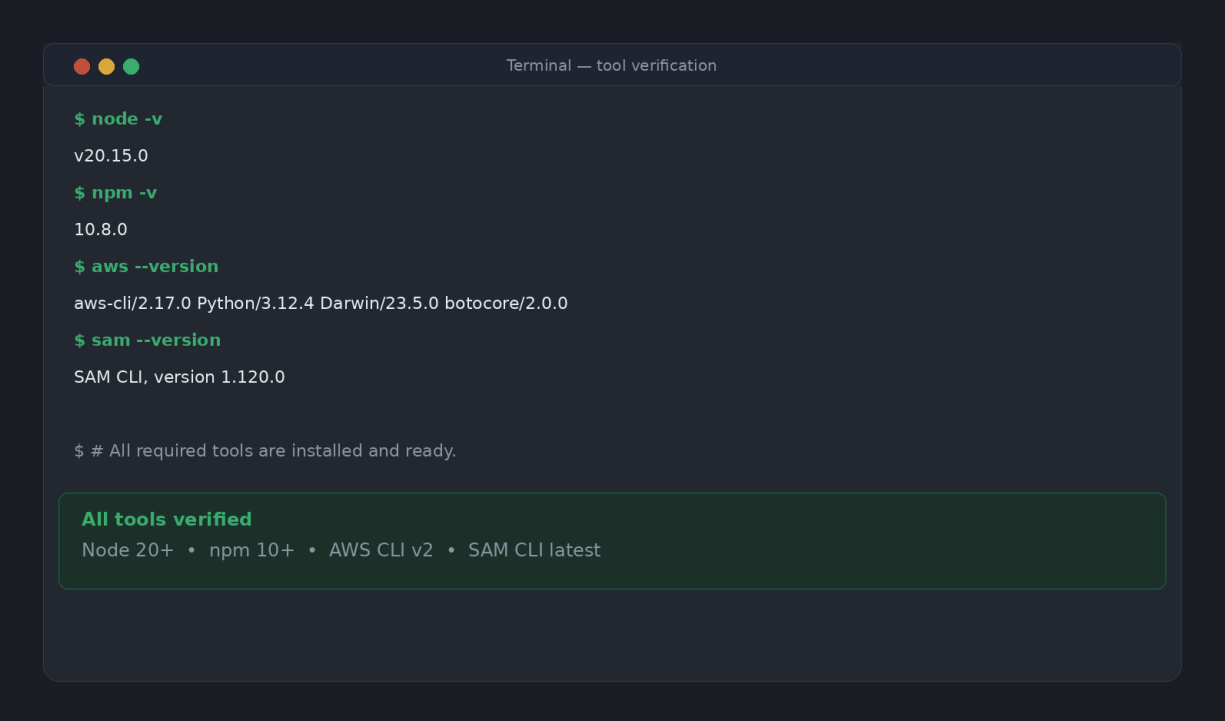
```
brew install awscli
```

Install AWS SAM CLI

```
brew install aws-sam-cli
```

Verify:

```
sam --version
```



```
Terminal — tool verification

$ node -v
v20.15.0
$ npm -v
10.8.0
$ aws --version
aws-cli/2.17.0 Python/3.12.4 Darwin/23.5.0 botocore/2.0.0
$ sam --version
SAM CLI, version 1.120.0

$ # All required tools are installed and ready.

All tools verified
Node 20+ • npm 10+ • AWS CLI v2 • SAM CLI latest
```

Tool verification in terminal

AWS Credentials

AWS Academy Lab Credentials

The project targets an AWS Academy student lab account. Lab credentials are temporary and expire each session.

1. Open the student lab portal.
2. Start the lab session.
3. Copy the credentials for the AWS CLI option.
4. Paste credentials into the root `.env` file in the repository:

```
AWS_ACCESS_KEY_ID=PASTE_HERE  
AWS_SECRET_ACCESS_KEY=PASTE_HERE  
AWS_SESSION_TOKEN=PASTE_HERE
```

5. Load credentials into the current shell:

```
set -a; source .env; set +a
```

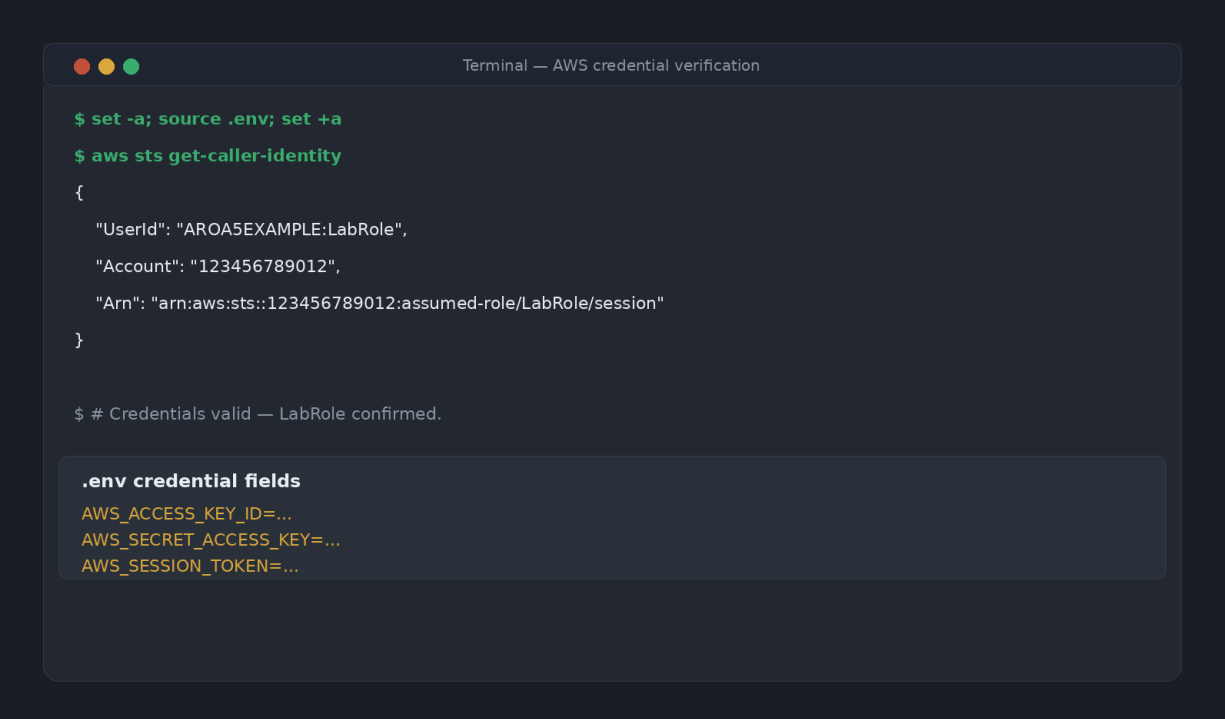
6. Verify credentials loaded correctly:

```
aws sts get-caller-identity
```

A successful response returns an account ID, user ID, and ARN. The ARN confirms the `LabRole` student identity.

Important rules:

- Never commit the root `.env` file. It is in `.gitignore`.
- Repeat the credential steps every new lab session because temporary credentials expire.
- Do not run `aws configure` with lab credentials. Use the `.env` load method instead.

A terminal window titled "Terminal — AWS credential verification" with a dark background. It shows the execution of two commands: "set -a; source .env; set +a" and "aws sts get-caller-identity". The output of the second command is a JSON object containing "UserId", "Account", and "Arn" fields. Below the output, a message states "# Credentials valid — LabRole confirmed." A separate box titled ".env credential fields" lists three environment variables: "AWS_ACCESS_KEY_ID=...", "AWS_SECRET_ACCESS_KEY=...", and "AWS_SESSION_TOKEN=...".

```
Terminal — AWS credential verification

$ set -a; source .env; set +a
$ aws sts get-caller-identity
{
  "UserId": "AROEXAMPLE:LabRole",
  "Account": "123456789012",
  "Arn": "arn:aws:sts::123456789012:assumed-role/LabRole/session"
}

$ # Credentials valid — LabRole confirmed.

.env credential fields
AWS_ACCESS_KEY_ID=...
AWS_SECRET_ACCESS_KEY=...
AWS_SESSION_TOKEN=...
```

AWS credential verification output

Deploy the Backend

First-Time Deployment

From the repository root:

```
npm install
npm run sam:build
npm run sam:deploy:guided
```

The guided deploy prompts for configuration. Use these values:

Prompt	Value
Stack name	salesops-ai-dev
Region	us-east-1
AWS profile	none
Confirm before deploy	Y
Allow IAM role creation	N
Disable rollback	N
Save config to file	Y

The template uses the AWS Academy LabRole for all Lambda functions. The deploy does not create new IAM roles because the lab environment blocks role creation.

After confirmation, SAM uploads the Lambda package and creates or updates the CloudFormation stack.

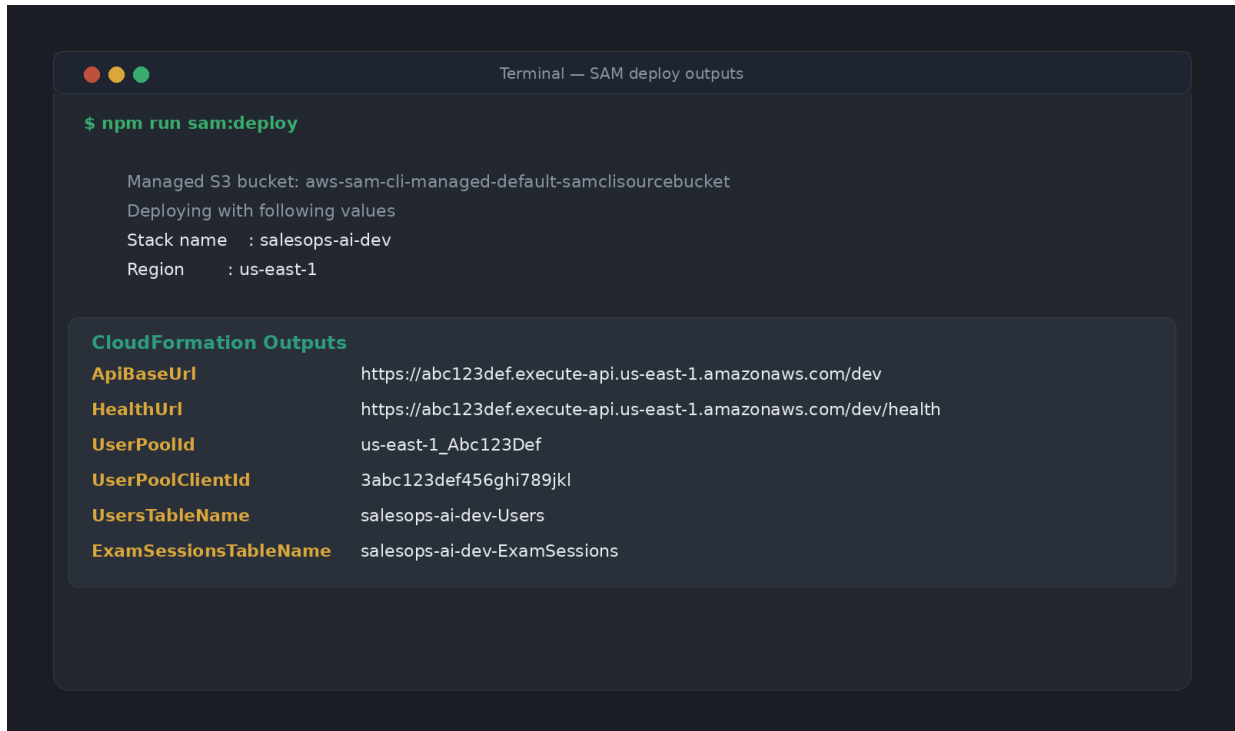
Deployment Outputs

After a successful deploy, SAM prints the stack outputs. Copy the ApiBaseUrl value. You will need it to configure the frontend.

Key outputs:

Output key	Description
ApiBaseUrl	Base URL for VITE_API_BASE_URL
HealthUrl	Health endpoint for smoke testing
UserPoolId	Cognito User Pool ID
UserPoolClientId	Cognito app client ID
UsersTableName	DynamoDB users table name
PersonasTableName	DynamoDB personas table name

Output key	Description
ScenariosTableName	DynamoDB scenarios table name
ExamSessionsTableName	DynamoDB exam sessions table name
ExamIssueReleaseQueueUrl	SQS queue URL

A terminal window titled "Terminal — SAM deploy outputs" showing the command "\$ npm run sam:deploy" and its output. The output includes a message about the managed S3 bucket, deployment values for stack name and region, and a section titled "CloudFormation Outputs" listing several key-value pairs.

```
Terminal — SAM deploy outputs

$ npm run sam:deploy

Managed S3 bucket: aws-sam-cli-managed-default-samclisourcebucket
Deploying with following values
Stack name   : salesops-ai-dev
Region      : us-east-1

CloudFormation Outputs
ApiBaseUrl   https://abc123def.execute-api.us-east-1.amazonaws.com/dev
HealthUrl    https://abc123def.execute-api.us-east-1.amazonaws.com/dev/health
UserPoolId   us-east-1_Abc123Def
UserPoolClientId 3abc123def456ghi789jkl
UsersTableName salesops-ai-dev-Users
ExamSessionsTableName salesops-ai-dev-ExamSessions
```

CloudFormation stack outputs screen

Re-Deployment

For subsequent deploys after the configuration file exists:

```
npm run sam:build
npm run sam:deploy
```

The non-guided deploy uses `backend/samconfig.toml` for all parameters. Review `samconfig.toml` before each deploy to confirm the stack name and region are correct.

Configure the OpenAI Secret

The issue generation and exam evaluation Lambda functions require an OpenAI API key stored in AWS Secrets Manager.

Create the Secret

Run once after the first deployment:

```
aws secretsmanager create-secret \
--name salesops/dev/llm-api-keys \
--secret-string '{"OPENAI_API_KEY":"PASTE_YOUR_KEY_HERE"}' \
--region us-east-1
```

Replace PASTE_YOUR_KEY_HERE with the actual OpenAI API key. Do not commit the key to version control.

Update an Existing Secret

If the secret already exists from a previous session:

```
aws secretsmanager put-secret-value \
--secret-id salesops/dev/llm-api-keys \
--secret-string '{"OPENAI_API_KEY":"PASTE_YOUR_KEY_HERE"}' \
--region us-east-1
```

Verify the Secret Exists

```
aws secretsmanager describe-secret \
--secret-id salesops/dev/llm-api-keys \
--region us-east-1
```

The command returns the secret ARN and name without revealing the key value.

AWS Management Console us-east-1

AWS Secrets Manager — salesops/dev/llm-api-keys

Secret name: **salesops/dev/llm-api-keys** Secret ARN: **arn:aws:secretsmanager:us-east-1:123456789012:secret:salesops/dev/llm-api-keys**

Secret value

```
{ "OPENAI_API_KEY": "sk-*****" }
```

Key is stored securely. Value is masked in the console.

AWS CLI command — create secret

```
aws secretsmanager create-secret \
  --name salesops/dev/llm-api-keys \
  --secret-string '{"OPENAI_API_KEY":"PASTE_YOUR_KEY_HERE"}' \
  --region us-east-1
```

Secret active **Never committed to version control**

Secrets Manager secret configuration

Behavior when the secret is missing or invalid:

- Issue generation falls back to clearly marked demo issues.
- Exam evaluation returns an error and no coaching result is created.
- A warning message appears in the frontend when demo issues were used.

Configure the Frontend

Create the Frontend Environment File

Create `frontend/.env.local` in the repository. This file is in `.gitignore` and must not be committed.

```
VITE_API_BASE_URL=https://YOUR_API_ID.execute-api.us-east-1.amazonaws.com/dev
```

Replace `YOUR_API_ID` with the `ApiBaseUrl` value from the deployment outputs.

Start the Development Server

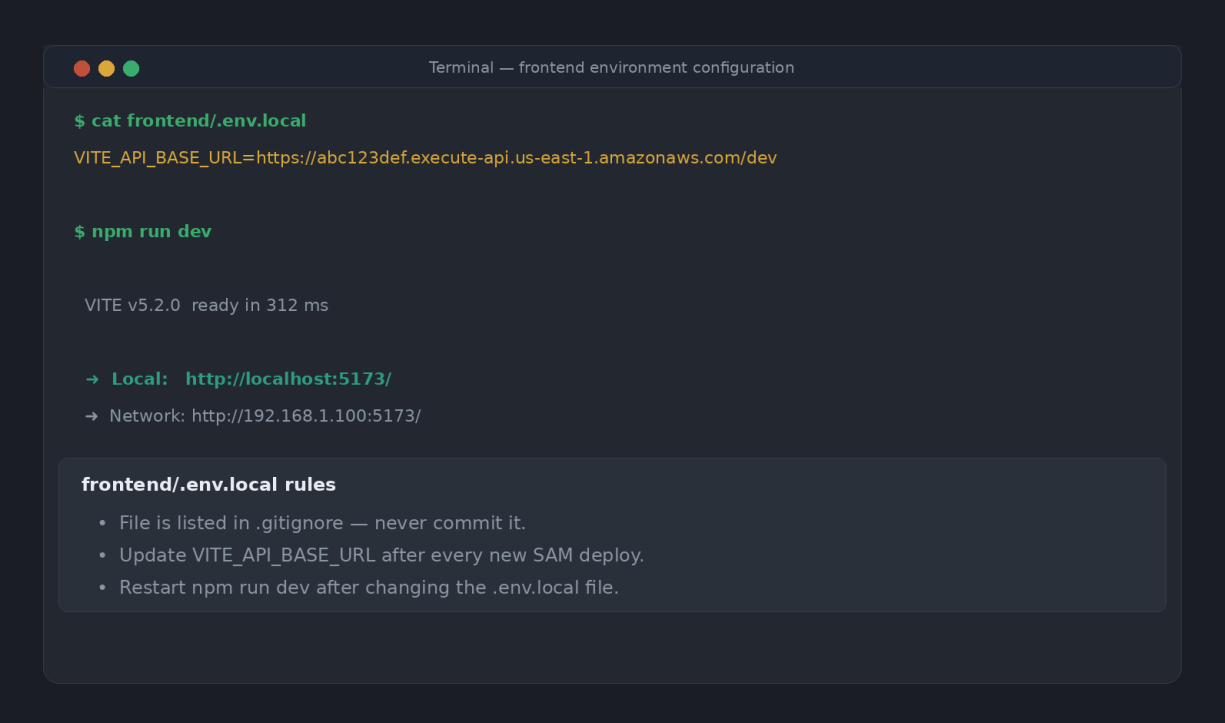
```
npm run dev
```

The frontend starts on `http://localhost:5173` by default. Open the address in a browser to confirm the login screen loads.

Build for Production

```
npm run build
```

The build output goes to `frontend/dist/`. Deploy this directory to a static hosting service such as Amazon S3 with Amazon CloudFront if a public deployment is required. Configure the S3 bucket and CloudFront distribution to serve `index.html` for all routes.

A terminal window titled "Terminal — frontend environment configuration" showing the following commands and output:

```
$ cat frontend/.env.local
VITE_API_BASE_URL=https://abc123def.execute-api.us-east-1.amazonaws.com/dev

$ npm run dev

VITE v5.2.0 ready in 312 ms

→ Local: http://localhost:5173/
→ Network: http://192.168.1.100:5173/

frontend/.env.local rules
• File is listed in .gitignore — never commit it.
• Update VITE_API_BASE_URL after every new SAM deploy.
• Restart npm run dev after changing the .env.local file.
```

Frontend environment configuration

Smoke Test

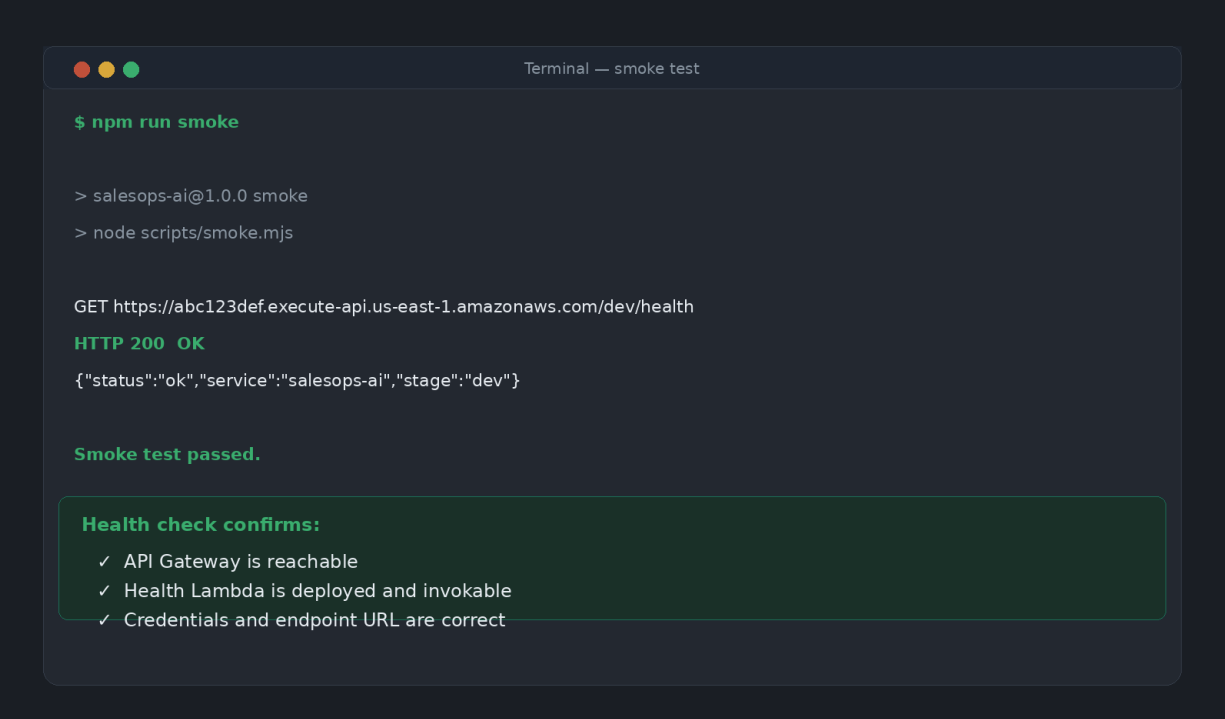
After deploy and frontend configuration, run the smoke test to confirm the API endpoint is reachable:

```
npm run smoke
```

The smoke script calls GET /health on the deployed API and verifies a successful response. A passing result confirms:

- API Gateway is running.
- The health Lambda function is deployed and invokable.
- Credentials are valid and the endpoint is reachable.

If the smoke test fails, check that credentials are loaded, the stack deployed successfully, and VITE_API_BASE_URL points to the correct endpoint.

A terminal window titled "Terminal — smoke test" with a dark background. It shows the command "\$ npm run smoke" and its output. The output includes the command being run, the HTTP status "200 OK", and a JSON response. A green box highlights the "Smoke test passed." message and a list of three confirmation points.

```
Terminal — smoke test

$ npm run smoke

> salesops-ai@1.0.0 smoke
> node scripts/smoke.mjs

GET https://abc123def.execute-api.us-east-1.amazonaws.com/dev/health
HTTP 200 OK
{"status":"ok","service":"salesops-ai","stage":"dev"}

Smoke test passed.

Health check confirms:
✓ API Gateway is reachable
✓ Health Lambda is deployed and invokable
✓ Credentials and endpoint URL are correct
```

Smoke test passing output

First Admin User Setup

After deployment, all newly registered users start as representatives with `PENDING_CONFIRMATION` status. There is no default admin account.

Step 1: Create a User Account

Open the application in a browser, create a new account, and confirm the email.

Step 2: Find the Cognito Sub

Look up the user record by email to get the Cognito sub (user ID):

```
aws dynamodb query \
--table-name salesops-ai-dev-Users \
--index-name EmailIndex \
--key-condition-expression "emailLower = :email" \
--expression-attribute-values '{"email":{"S":"admin@example.com"}}' \
--region us-east-1
```

Copy the `userId` value from the result. This is the Cognito sub.

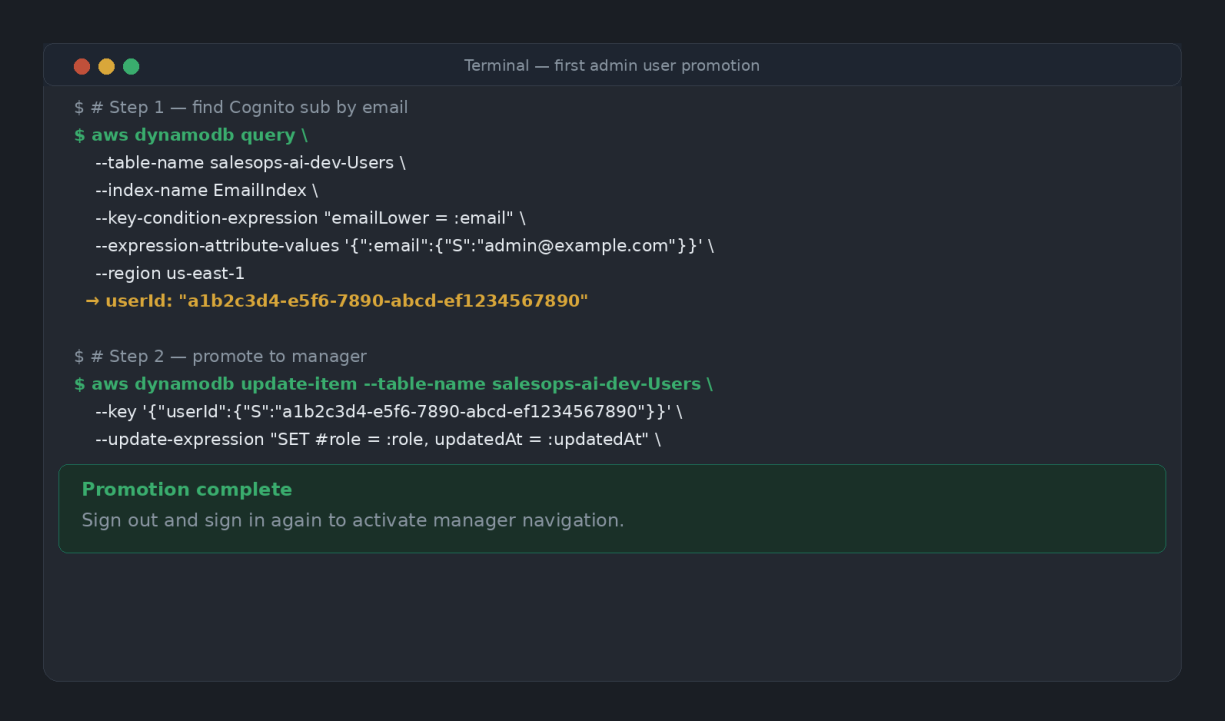
Step 3: Promote to Manager

```
aws dynamodb update-item \
--table-name salesops-ai-dev-Users \
--key '{"userId":{"S":"PASTE_USER_ID_HERE"}}' \
--update-expression "SET #role = :role, updatedAt = :updatedAt" \
--expression-attribute-names '{"#role":"role"}' \
--expression-attribute-values \
'{"role":{"S":"manager"},"updatedAt":{"S":"2026-05-27T00:00:00.000Z"}}' \
--region us-east-1
```

Replace `PASTE_USER_ID_HERE` with the Cognito sub from step 2.

Step 4: Verify Promotion

Sign out of the application and sign in again. The sidebar should show the manager navigation items: Content Library, Personas, Scenarios, Dashboard, and Users.

A terminal window titled "Terminal — first admin user promotion" with three colored window control buttons (red, yellow, green) in the top-left corner. The terminal displays two steps of a process. Step 1 is a query to find a user by email, returning a specific user ID. Step 2 is an update command to promote that user to a manager role. A green box at the bottom indicates the promotion is complete and provides instructions to sign out and back in.

```
Terminal — first admin user promotion

$ # Step 1 — find Cognito sub by email
$ aws dynamodb query \
  --table-name salesops-ai-dev-Users \
  --index-name EmailIndex \
  --key-condition-expression "emailLower = :email" \
  --expression-attribute-values '{":email":{"S":"admin@example.com"}}' \
  --region us-east-1
→ userId: "a1b2c3d4-e5f6-7890-abcd-ef1234567890"

$ # Step 2 — promote to manager
$ aws dynamodb update-item --table-name salesops-ai-dev-Users \
  --key '{"userId":{"S":"a1b2c3d4-e5f6-7890-abcd-ef1234567890"}}' \
  --update-expression "SET #role = :role, updatedAt = :updatedAt" \

Promotion complete
Sign out and sign in again to activate manager navigation.
```

DynamoDB admin user promotion

After the first manager account exists, that manager can promote other users from the Users page inside the application without needing AWS CLI access.

DynamoDB Administration

View Table Contents

List items in the Users table:

```
aws dynamodb scan \
--table-name salesops-ai-dev-Users \
--region us-east-1
```

List items in the Personas table:

```
aws dynamodb scan \
--table-name salesops-ai-dev-Personas \
--region us-east-1
```

Suspend a User by CLI

```
aws dynamodb update-item \
--table-name salesops-ai-dev-Users \
--key '{"userId":{"S":"COGNITO_SUB"}}' \
--update-expression "SET #status = :status, updatedAt = :updatedAt" \
--expression-attribute-names '{"#status":"status"}' \
--expression-attribute-values \
'{"":status":{"S":"SUSPENDED"},":updatedAt":{"S":"2026-05-27T00:00:00.000Z"}}' \
--region us-east-1
```

Clear All Exam Sessions

Use only in development or testing scenarios. This deletes all exam data and cannot be undone.

```
aws dynamodb scan \
--table-name salesops-ai-dev-ExamSessions \
--attributes-to-get sessionId recordId \
--region us-east-1
```

Delete items individually using the returned keys. There is no bulk delete command in the AWS CLI. Use the AWS Console table editor or a custom script for bulk operations.

Cognito Administration

View the User Pool in the Console

1. Open the AWS Management Console.
2. Navigate to Amazon Cognito.
3. Select the User Pool named salesops-ai-dev-users.

The console shows all registered users, their confirmation status, and last sign-in time.

Delete a User from Cognito

Deleting a user from Cognito does not delete the DynamoDB profile. Delete both records if a full removal is needed.

From the Cognito console, select the user and choose Delete. To use the CLI:

```
aws cognito-idp admin-delete-user \  
--user-pool-id YOUR_USER_POOL_ID \  
--username user@example.com \  
--region us-east-1
```

Get the UserPoolId from the CloudFormation stack outputs.

Reset a User Password by Admin

```
aws cognito-idp admin-reset-user-password \  
--user-pool-id YOUR_USER_POOL_ID \  
--username user@example.com \  
--region us-east-1
```

The user receives an email with a temporary password.

Routine Maintenance

Refresh Lab Credentials

Lab credentials expire. Before each new lab session:

- 1. Open the lab portal and start the session.
- 2. Copy the new credentials.
- 3. Paste into the root .env file.
- 4. Run `set -a; source .env; set +a`.
- 5. Verify with `aws sts get-caller-identity`.

Rebuild and Redeploy After Code Changes

After modifying any Lambda handler:

```
npm run sam:build
npm run sam:deploy
npm run smoke
```

Confirm the smoke test passes after each deployment.

Check Stack Status

```
aws cloudformation describe-stacks \
--stack-name salesops-ai-dev \
--region us-east-1 \
--query "Stacks[0].StackStatus"
```

Expected values:

Status	Meaning
CREATE_COMPLETE	First deployment succeeded
UPDATE_COMPLETE	Update deployment succeeded
ROLLBACK_COMPLETE	Deployment failed and rolled back
UPDATE_ROLLBACK_COMPLETE	Update failed and rolled back

View Stack Events After a Failed Deployment

```
aws cloudformation describe-stack-events \
--stack-name salesops-ai-dev \
--region us-east-1 \
--query "StackEvents[?ResourceStatus=='CREATE_FAILED' || ResourceStatus=='UPDATE_FAILED']"
```


Review the ResourceStatusReason field for the root cause.

Delete the Stack

To remove all infrastructure:

```
aws cloudformation delete-stack \  
--stack-name salesops-ai-dev \  
--region us-east-1
```

This deletes all Lambda functions, API Gateway, Cognito resources, DynamoDB tables, and the SQS queue. DynamoDB data is permanently lost. Confirm with the team before running this command.

CloudWatch Logs

Lambda function logs are written to CloudWatch Logs automatically. Each function has its own log group named `/aws/lambda/<FunctionName>`.

View Recent Logs for the Auth Functions

```
aws logs tail /aws/lambda/salesops-ai-dev-SignUpFunction \
--follow \
--region us-east-1
```

Replace the function name with any deployed Lambda name from the CloudFormation stack.

List All Log Groups

```
aws logs describe-log-groups \
--log-group-name-prefix /aws/lambda/salesops-ai \
--region us-east-1
```

Find Errors in Logs

```
aws logs filter-log-events \
--log-group-name /aws/lambda/salesops-ai-dev-GenerateScenarioIssuesFunction \
--filter-pattern ERROR \
--region us-east-1
```

Use this to diagnose issue generation or evaluation failures without opening the AWS console.

AWSManagement Consoleus-east-1

Amazon CloudWatch — Lambda log groups

Log group name	Retention	Stored bytes	Last event
/aws/lambda/salesops-ai-dev-SignUpFunction	Never	142 KB	2 min ago
/aws/lambda/salesops-ai-dev-SignInFunction	Never	88 KB	5 min ago
/aws/lambda/salesops-ai-dev-GenerateScenarioIssuesFunction	Never	1.2 MB	12 min ago
/aws/lambda/salesops-ai-dev-CreateExamEvaluationFunction	Never	3.4 MB	18 min ago
/aws/lambda/salesops-ai-dev-GetDashboardFunction	Never	310 KB	25 min ago
/aws/lambda/salesops-ai-dev-ReleaseExamIssueFunction	Never	56 KB	31 min ago

View live logs via CLI

aws logs tail /aws/lambda/salesops-ai-dev-GenerateScenarioIssuesFunction \
--follow --region us-east-1

CloudWatch log monitoring

Troubleshooting

Smoke Test Fails with Connection Error

- Confirm lab credentials are loaded: `aws sts get-caller-identity`.
- Confirm the `VITE_API_BASE_URL` in `frontend/.env.local` matches the `ApiBaseUrl` stack output.
- Check the CloudFormation stack status. If it shows `ROLLBACK_COMPLETE`, the deployment did not succeed.

SAM Build Fails

- Confirm `node -v` returns 20 or newer.
- Run `npm install` from the repo root.
- Check for syntax errors in modified Lambda handler files.
- Delete the `.aws-sam/` build cache directory and retry.

SAM Deploy Fails with Credential Error

- Lab credentials may have expired. Refresh the `.env` file and reload.
- Confirm the region is `us-east-1` in `samconfig.toml`.

CloudFormation Stack Stuck in ROLLBACK

- Run `aws cloudformation describe-stack-events` to read the failure reason.
- Fix the root cause in `template.yaml` or Lambda handler code.
- If the stack is in `ROLLBACK_COMPLETE` state, delete and redeploy from scratch.

Users Cannot Sign In After Redeploy

- Check the Cognito User Pool ID in the new stack outputs.
- If the User Pool was deleted and recreated, all user accounts are gone. Users must re-register.
- Cognito User Pools are not replaced unless the SAM template resource name changes.

Issue Generation Returns Demo Issues

- Confirm the Secrets Manager secret exists: `aws secretsmanager describe-secret --secret-id salesops/dev/llm-api-keys`.
- Confirm the secret value contains a valid `OPENAI_API_KEY`.
- Check CloudWatch logs for the `GenerateScenarioIssuesFunction` log group for error details.

Lambda Timeout on Evaluation

- Exam evaluation uses a 30-second timeout and 256 MB memory.

- If timeouts occur frequently, check CloudWatch for the failure pattern.
- In the lab environment, OpenAI response time can vary. Demo mode is not available for evaluation.

SQS Issues Not Releasing on Time

- The `ReleaseExamIssueFunction` consumes SQS messages with a delay.
- The pulse endpoint also reveals due issues by comparing `releaseAt` with the current time.
- If no issues appear during an exam, confirm the exam session started and the pulse endpoint is responding.

Security Notes

- Do not commit `.env`, `.env.local`, or any file containing AWS credentials or API keys.
- The `.gitignore` file already excludes these files.
- CORS in the current template uses a wildcard origin (*). Replace with the final frontend origin before any public deployment.
- All Lambda functions use the shared AWS Academy LabRole. For non-lab production deployment, replace with least-privilege IAM policies for each function.
- Cognito User Pool passwords require a minimum of 8 characters, one uppercase letter, one lowercase letter, and one number.
- Representatives can only access their own exam sessions. The backend enforces session ownership checks on every exam route.
- Managers cannot demote or suspend their own accounts. The backend blocks this action.

Configuration Reference

SAM Template Parameters

Parameter	Default	Description
StageName	dev	API Gateway stage and resource name suffix
LlmSecretName	salesops/dev/llm-api-keys	Secrets Manager secret name for OpenAI key
OpenAiModel	gpt-5-mini	OpenAI model for issue generation and evaluation

Lambda Timeouts and Memory

Function	Timeout	Memory	Reason
All others	10 s	128 MB	Default
GenerateScenarioIssuesFunction	25 s	256 MB	OpenAI call
CreateExamEvaluationFunction	30 s	256 MB	OpenAI call
GetDashboardFunction	15 s	256 MB	DynamoDB scan

DynamoDB Tables

Table	Partition key	Sort key	Notes
Users	userId	none	EmailIndex GSI on emailLower
Personas	personaId	none	
Scenarios	scenarioId	none	Issues stored inside the item
ExamSessions	sessionId	recordId	Composite key for META and ISSUE records

Cognito Token Validity

Token	Validity
Access token	1 hour
ID token	1 hour

Token	Validity
Refresh token	30 days

After an ID token expires, the frontend uses the refresh token to obtain a new one automatically.

Recommended Admin Checklist Per Session

1. Start the lab session and copy fresh credentials.
2. Load credentials: `set -a; source .env; set +a`.
3. Verify identity: `aws sts get-caller-identity`.
4. Confirm the CloudFormation stack is in `UPDATE_COMPLETE` or `CREATE_COMPLETE`.
5. Confirm the Secrets Manager secret exists for the OpenAI key.
6. Run the smoke test: `npm run smoke`.
7. Start the frontend: `npm run dev`.
8. Sign in as a manager to confirm the full application is working.